

CSE340: Computer Architecture

Assignment 4

Chapter 4

Question 01

Suppose you have a single instruction;

Add X21, X22, X23

Register read = 20s

Instruction Fetch = 20s

Memory Operation = 40s

Register write = 20s

ALU Operation = 40s

Suppose you have a pipelined setup and a single-cycle setup,
Which one will run the instruction faster?

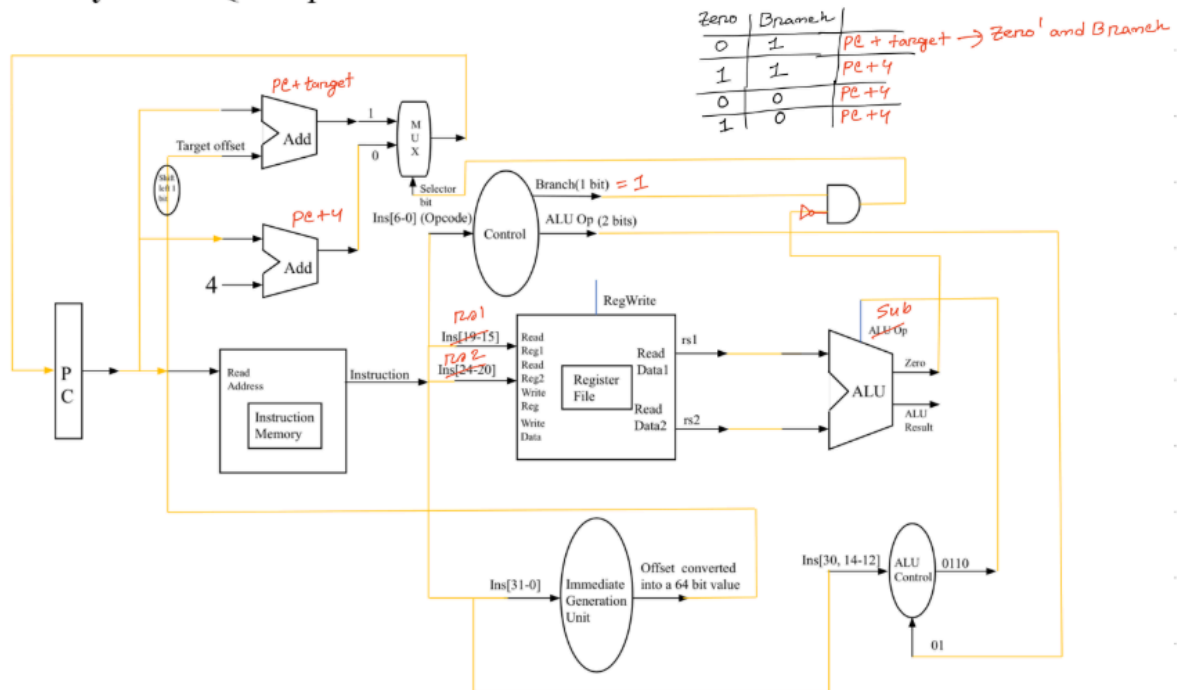
Add in single cycle = Instruction Fetch + Register Read + ALU Operation +
Register Write
 $= 20 + 20 + 40 + 20 = 100s$

Add in pipeline,
Highest stage/ delay = 40
So, $40 * 5 = 200s$

Single cycle will run faster.

Question 02

Modify the BEQ datapath so that it works for BNE.



Question 03

Explain control hazard in your own words.

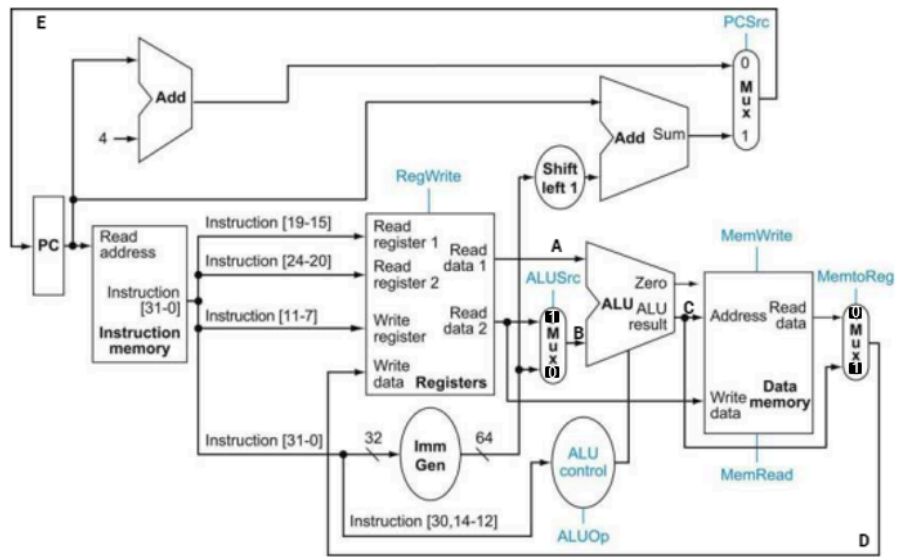
Do you own research.

Question 04

Why do we keep instruction and data memory separate?

Do you own research.

Question 05



Write the control signals for the following instruction:
Ld X5, 10(X21);

- i. ALU Src; **0**
- ii. RegWrite; **1**
- iii. MemRead; **1**
- iv. MemWrite; **0**
- v. MemtoReg. **0**